

Guidelight

Project report



Illustration generated with Artificial Intelligence

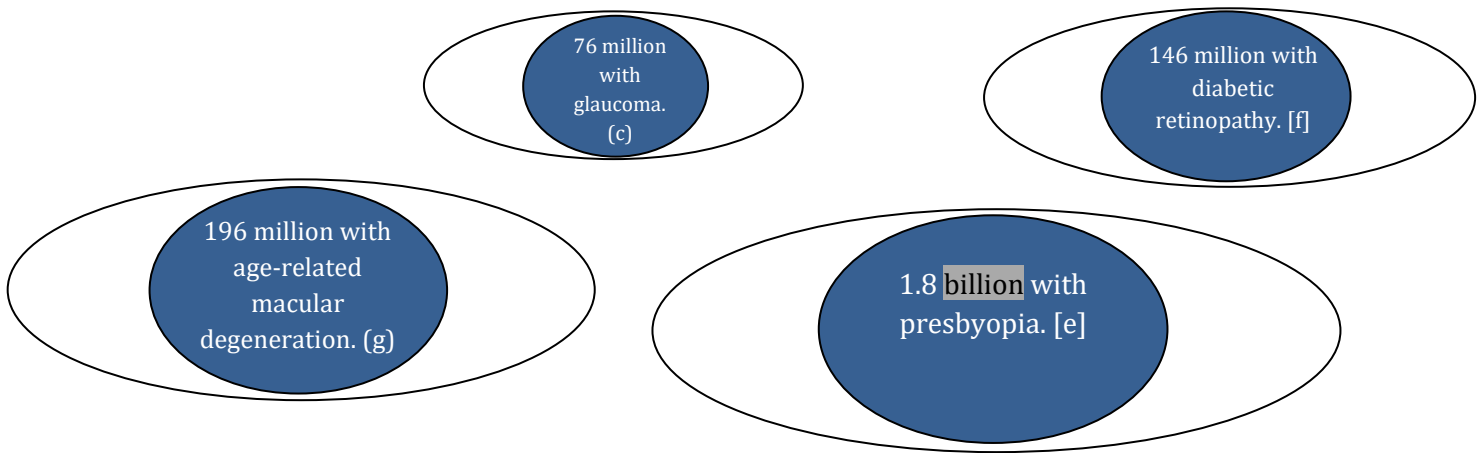
1. Problem Statement and Impact

1.1 Targeted Disability

Primary: People with visual impairments (Blind & Low-Vision).

Secondary: People with hearing impairments (Hearing Loss), including those who require hearing rehabilitation.

→ Globally, at least **2.2 billion** people have a vision impairment.



c: 76 million (95% credible intervals (CrI), 51.9–111.7) people (40 to 80 years of age) with glaucoma in 2020.

e: 1.8 billion (confidence interval [CI], 1.7–2.0) people of all ages with presbyopia in 2015.

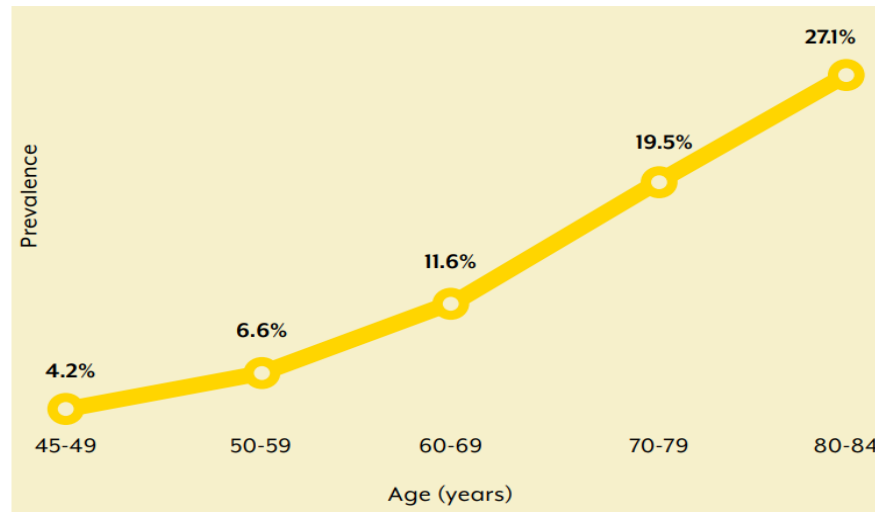
f: 146 million adults with diabetic retinopathy was calculated by applying the global prevalence of any diabetic retinopathy (34.6%) reported by Yau et al. [2012] to the estimated global number of adults aged over 18 years of age with diabetes in 2014 (422 million) that was reported in the WHO Global Report on Diabetes, 2016.

g: 195.6 million (95% CrI 140–261) people aged 30 to 97 years with age-related macular degeneration in 2020

Within this group, one of the most relevant and critical segments is elderly people. **Age** plays a significant role in determining the target users of the device for two main reasons:

1. High prevalence of vision loss among the elderly:

Partial or severe vision loss is far more common in older adults than in younger individuals. Unlike younger people, whose vision problems can often be corrected with glasses, contact lenses, or medical procedures, many elderly individuals suffer from **irreversible vision loss** due to conditions such as *glaucoma*, *macular degeneration*, or *diabetic retinopathy*. These conditions cannot be solved by simple corrective tools, making assistive technologies such as smart crutches highly relevant for them.



The overall prevalence of age-related macular degeneration is estimated to increase 7-fold, from 4.2% in those aged 45–49 years, to 27.2% in those aged 80–85 years

2. Social and cultural association of crutches with older age:

While younger individuals with visual impairment may adapt to alternative solutions or surgical interventions, **crutches are naturally associated with older adults**. It is culturally and socially uncommon for a young person to walk with a crutch unless there is a severe disability. For older adults, however, the use of a crutch feels more practical, acceptable, and expected.

By focusing on elderly users, the device ensures **higher adoption rates**, since it responds to both a medical reality (irreversible vision loss) and a social reality (acceptance of mobility aids in older age).

1.2 Market Gap Analysis

Assistive Technology (AT) Market Size 2025-2029

The Assistive Technology (AT) market size is forecast to **increase by USD 6.3 billion**. at a CAGR of **4.5%** between **2024** and **2029**.

The market is experiencing **significant growth** due to the rising number of orthopedic and neurological disorders. This trend is driven by an aging population and increasing prevalence of chronic conditions, leading to a greater demand for mobility and communication solutions. Furthermore, growing consumer awareness of health and wellness in urban areas is fueling the adoption of AT devices, particularly among the elderly and disabled populations.

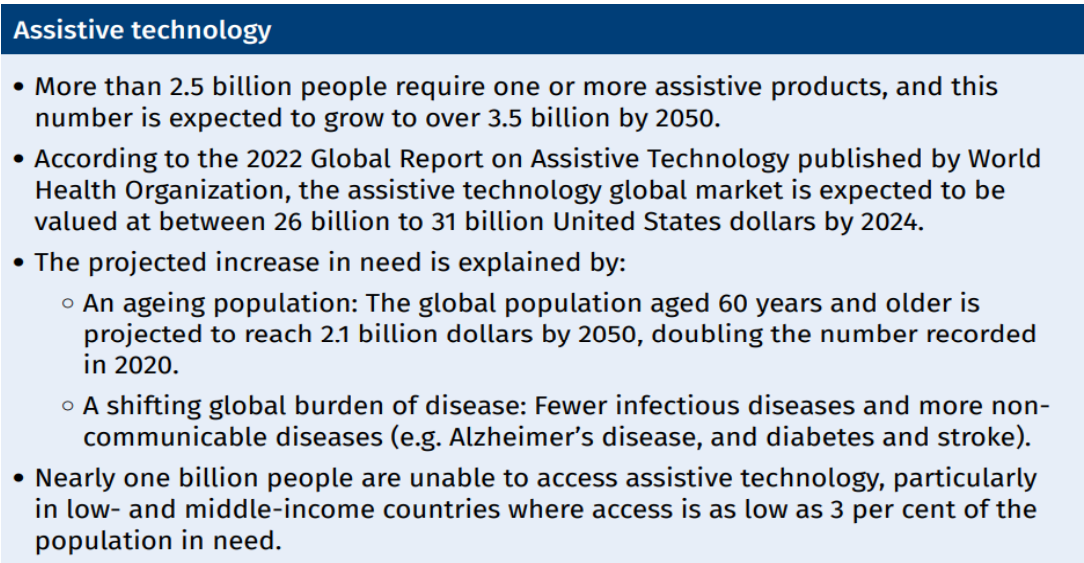


Figure 1. Key facts and projections about the global assistive technology market.

The Lancet Global Health Commission on Global Eye Health estimates that approximately **43.3 million people were blind in 2020**, and nearly **295** million people had moderate and severe vision impairment (**MSVI**)¹.

Nearly 55% were women, and 70% were **older than 50 years**. There are also significant regional variations in prevalence. Globally, 92% of people who are blind and 88% of those with MSVI live in LMICs. Western sub-Saharan Africa has the highest age-standardized prevalence of blindness (1.1%), and North America the lowest (0.1 %). South Asia has the highest age standardized prevalence of MSVI (6.4%). Because of large regional populations, the highest number of blind people live in South Asia (11.9 million) and East Asia (9.1 million).

COVID-19 led to a growing recognition worldwide about the importance of assistive technologies accessible to all. The market for screen readers is poised for growth due to:

1 Moderate and severe vision impairment is defined as visual acuity worse than 6/18, but equal to or better than 3/60. Blindness is defined as visual acuity worse than 3/60.

- **Growing need & demand:** The high prevalence of vision loss/blindness is a significant factor driving market growth. Additionally, the expanding ageing populations, associated with various vision-related disorders, contributes to the demand for assistive technologies.
- **Artificial intelligence (AI):** Advances in machine learning and AI have enhanced screen reader capabilities, improving accuracy and flexibility in response to evolving digital content forms. Devices are no longer just tools—they're becoming **intelligent companions** that can adapt to the user's environment, learn patterns, and provide proactive assistance (e.g., fall prediction, obstacle classification, personalized alerts).
- **User-Centered & Inclusive Design:** AT design is shifting from a “medical device” perspective to a **human-centered approach**. Devices are expected to be comfortable, discreet, stylish, and **customizable**—not just functional. This helps reduce stigma and encourages daily use.

1.3 Proposed Solution

Guide-Light: A smart cane equipped with a camera + sensors + integrated earpiece, with on-device processing for obstacle detection, plus **sensor fusion** for real-time distance/risk estimation.

It provides:

- **Natural voice guidance (TTS) or vibration feedback**, depending on user profile (vision impairment only, hearing impairment only, or both).
- **Companion family app:** Alerts in case of danger/assistance request, optional location sharing, and privacy-controlled activity updates.

1.4 Opportunities (Impact)

Where does Guidelight stand? (why choose ours?)

The smart crutch system is expected to create a **transformational impact** on individuals with disabilities, especially elderly people. Its benefits extend across multiple dimensions:

1. **Mobility:** The device helps elderly users navigate safely in their surroundings, reducing the risk of accidents and allowing them to move around with greater ease.
2. **Independence:** the smart crutch offers a way to move independently without constant reliance on family members or caregivers. This reduces their dependency and gives them back a sense of freedom in daily life.
3. **Psychological well-being:** Loss of vision in old age often leads to feelings of helplessness, social isolation, and reduced confidence. By enabling safe and reliable navigation, the smart crutch restores dignity and self-esteem.
4. **Family and caregiver relief:** Since elderly individuals often depend heavily on family members for mobility, it reduces the emotional and physical burden on caregivers.

By targeting the elderly demographic, the smart crutch addresses a **critical gap** in assistive technologies: it does not simply correct vision, but instead provides **a lifelong supportive tool** for those who cannot regain their sight through medical means.

2. Literature Survey and Market Research

2.1 Existing Solutions Today

1. **We-WALK Smart Cane 2:** Cane with sonar + smartphone/voice assistant integration.



We-WALK Smart Cane

Current pricing: **\$850** with voice assistant subscription **\$4.99/month**, or **\$1150 all-inclusive**. Some vendors list it at around **\$1050**.

2. **Dual-beam sonar cane:** priced around **£621–£635** in the UK. (ultracane.com, livingmadeeasy.org.uk)
3. **Sunu Band:** smart mobility wristband that uses ultrasonic technology to detect and alert you to navigational obstacles up to 14 ft. away; which is discontinued now. (aph.org)
4. **Smartphone features:** iPhone (Magnifier/Detection Mode: Door & People Detection) but requires **LiDAR** on Pro models only. ([Apple Support](https://support.apple.com), [Apple](https://apple.com)).



Dual-beam sonar



Sunu Band

2.1 General Gaps Across All Devices

1. **Accessibility of the Companion App** – Most apps assume some smartphone literacy. Seniors and newly visually impaired users struggle with setup and daily usage.
2. **Health & Safety Integration** – None of these solutions connect strongly with healthcare apps (fall detection, heart rate, daily activity tracking).
3. **Family/Community Connection** – Limited or no way for family members to get alerts, activity logs, or check user safety (unless through third-party apps).
4. **Future-proofing** – Updates are limited; most don't have flexible OTA upgrade models that continuously improve AI or sonar capabilities.
5. **Inclusive Design** – Many assume only young/mid-age tech-savvy users, ignoring older demographics who need simpler interfaces.

3. Methods and Appropriateness of Solution

3.1 Hardware methods

1. **Ultrasonic sensors:** an instrument that measures the distance to an object using **ultrasonic sound waves**. One of the best ways to sense proximity and detect levels with high reliability. Used for short-range obstacle detection because they are low-cost, reliable, and accurate in measuring distances up to a few meters. They improve the usability of the camera by also detecting objects that might be **transparent** or difficult to capture visually like glasses or walls.

2. **Camera + YOLO model:** Helps with instant, real-time objects detection and classification. The model only looks at the image once and from this 'single pass' is able to identify objects in the image. Which is a fast, lightweight, and optimized for embedded systems, allowing real-time recognition even on limited hardware.
3. **Microcontroller (Raspberry Pi or Arduino):** Raspberry Pi is ideal for running the YOLO model and managing complex tasks (camera input, sensor fusion, voice output). Arduino can be used as a lightweight controller for simple sensor readings. This hybrid choice ensures efficiency.
4. **Earpiece (bone conduction):** Bone conduction is especially appropriate for visually impaired individuals because it leaves ears open for environmental sounds, enhancing safety.

3.2 Software methods

1. **Embedded system for sensor fusion:** Ensures integration of multiple inputs (camera + ultrasonic sensors) to provide accurate and reliable obstacle detection.
2. **YOLO (object detection):** Appropriate because the algorithm (You Only Look Once) can classify multiple objects in **real time**, which is critical especially for guiding users in dynamic outdoor environments.
3. **Voice guidance system (Text-to-Speech):** Converts system decisions into understandable instructions for the user. Voice prompts are natural, and the option of **vibration** feedback supports users with hearing impairments.

3.3 Appropriateness

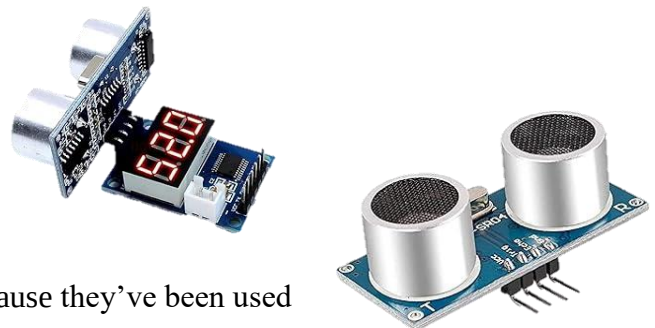
While our goal was to balance **accuracy**, **speed**, **cost**, and **usability**, our system—using the YOLO model and the mentioned sensors—ensures accurate detection across different scenarios. The hardware choices (Raspberry Pi, bone conduction earpiece) are **feasible** for portability and user comfort.

4. Resources Needed

4.1 Hardware Resources

1. Sensors:

Ultrasonic modules (e.g., HC-SR04) are easily found on electronics marketplaces like **Amazon**, **eBay**, **AliExpress**, **Spark Fun**, **Adafruit**, or local electronics shops. They're also stocked in university labs, makerspaces, and hardware prototyping labs. Because they've been used for decades in robotics, automation, and even cars (for parking assistance), they are **not hard to source**.

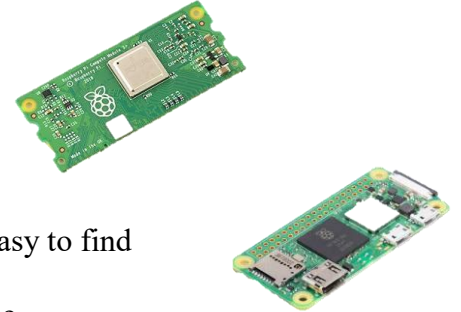


2. Camera:

Raspberry Pi camera modules are affordable and well-documented, making them easy to integrate.

3. Microcontroller:

Raspberry Pi boards are available in most electronics shops and online. Arduino boards are even cheaper and accessible.



Current Availability (as of 2025):

- **Raspberry Pi 4 Model B** → still the most commonly used; usually easy to find in electronics shops and online.
- **Raspberry Pi 5** → released in late 2023, it's becoming more available worldwide, but demand is still very high, so sometimes you'll see limited stock.
- **Raspberry Pi Zero 2 W** → very compact and cheaper, but often sells out quickly because it's popular for embedded and IoT projects.

4. Earpiece:

Bluetooth and **bone conduction** headsets are commercially available and compatible with Raspberry Pi. **Bone conduction** is a technology that converts sound into mechanical vibrations transmitted through the skin and temporal bone to the cochlea, bypassing traditional air conduction transmitters like the external auditory canal or eardrum.



4.2 Software Resources

1. **YOLOv8 model:** Open-source and well-documented; multiple lightweight pre-trained versions (YOLOv5n, YOLOv7-tiny) are available.
2. **Embedded software tools:** Python, OpenCV, TensorFlow/PyTorch are open-source and free, with strong community support.
3. **Voice guidance (TTS):** Open-source libraries (e.g., eSpeak, pyttsx3) can be easily integrated.

→ All resources (HW & SW) are **low-cost, open-source, and widely available**.

This ensures feasibility within academic or early-stage prototyping environments.

5. Budget Breakdown

Item	Estimated Cost (USD)
Raspberry Pi 4 (base)	\$35 (base 1GB model) (Wikipedia)
Ultrasonic Sensors ×2	\$6.99 for 2-pack (each ~\$3.50) (Amazon)
Bluetooth Earbuds	\$15–\$20 (budget earbuds on Amazon) (Amazon)
Camera Module	Assumed ~\$15 (common Pi-compatible camera)
Power Bank	\$10 (standard estimate)
Miscellaneous Components	\$20 (wires, PCB, 3D printing, enclosure)

→ **Total** = Approximately **\$102**

6. SWOT Analysis

Strengths	Weaknesses
• Combines sensors and AI for accurate obstacle detection. • Real-time voice guidance improves safety. • Portable and user-friendly design. • Addresses both visual and hearing impairments.	• Higher cost compared to traditional canes. • Requires regular charging (limited battery life) • Dependence on technical expertise for maintenance. • Initial learning curve for new users.
Opportunities	Threats
• Growing demand for assistive technologies. • Integration with mobile apps for remote monitoring. • Potential government and NGO support for accessibility projects. • Expanding market in aging populations worldwide.	• Competition from existing products like WeWALK and Sunu Band. • Rapid technological changes may outdate the solution. • Affordability challenges in low-income regions. • Privacy/security concerns with camera-based systems.

7. Time Plan and Deliverables

Week	Task / Deliverable
1	Literature Review and Market Research
2	System Design (Hardware & Software Architecture)
3	Hardware Setup (Raspberry Pi, Camera, Sensors, Earphone)
4	Software Development (Sensor integration, Obstacle detection logic)
5	System Integration (Hardware + Software testing)
6	Prototype Testing with Users (Feedback & Iteration)
7	Documentation Preparation (Report, Presentation)
8	Final Review and Submission

9. Business Model

- **Proposition:** Affordable smart crutch for visually impaired.
- **Customer segments:** Blind associations, NGOs, hospitals, individuals.
- **Revenue streams:** Device sales, maintenance +

Future expansions:

In addition to the current system (crutch and earpiece), new assistive devices can be integrated in the future, such as screen readers and other accessibility tools. These future developments will also contribute to revenue growth.

- **Cost structure:** Manufacturing, sensors, assembly.
- **Channels:** Online sales, NGOs, hospitals.
- **Key partners:** Tech companies, NGOs, government health departments.